

Quiz Performance Report

Performance Summary

A detailed breakdown of your quiz results and performance

Total Questions

30

Questions Attempted

30

Questions Passed

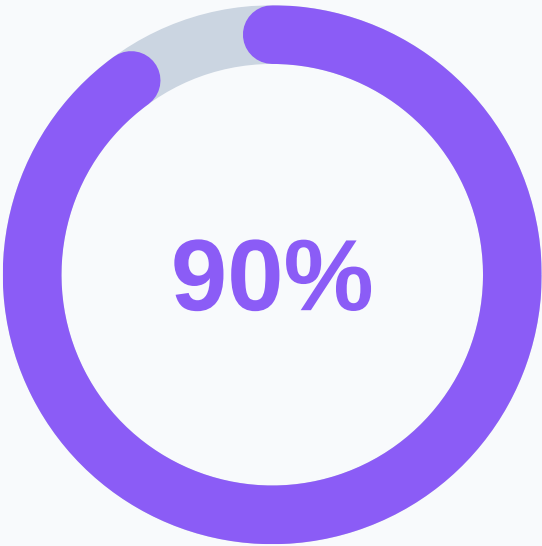
27

Questions Failed

3

Overall Score

Your overall quiz score percentage



Quiz Questions

1. Your REST API is returning 500 error randomly under load. What is the most likely reason?

EASY

Frontend bug

Server overload or exception

Correct Answer

Database deleted

HTML issue

Explanation

Unhandled exceptions or thread/resource exhaustion.

2. Why should you use DTO instead of directly exposing entity in API?

EASY

Security & abstraction

Correct Answer

Faster

Less memory

Easier coding

Explanation

Prevents data leakage and improves control.

3. What happens if @Transactional is missing in a multi-step DB operation?

EASY

Faster

Data inconsistency

Correct Answer

Compile error

No issue

Explanation

Partial updates may occur.

4. Why is indexing important in database queries?

EASY

Reduces memory

Deletes data

Faster lookup

Correct Answer

Avoids joins

Explanation

Speeds up search operations.

5. What issue occurs when API response time increases with data size?

EASY

UI issue

Network issue

Your Answer

Query inefficiency

Correct Answer

CSS problem

Explanation

Poor query or no pagination.

6. Why should pagination be used in APIs?

EASY

UI improvement

Performance optimization

Correct Answer

Security

Faster CSS

Explanation

Prevents loading huge data at once.

7. What happens if frontend sends wrong JSON structure?

EASY

Error

Correct Answer

Works

Ignored

Auto fix

Your Answer

Explanation

Deserialization failure.

8. What is the best way to handle global exceptions in Spring Boot?

EASY

Try-catch everywhere

Ignore

Print logs

@ControllerAdvice

Correct Answer

Explanation

Use @ControllerAdvice.

9. Why is connection pooling important?

EASY

Reduce overhead

Correct Answer

Faster UI

More memory

Security

Explanation

Reuses DB connections.

10. What issue arises if you fetch all data and filter in Java instead of DB?

EASY

Faster

Secure

Memory waste

Correct Answer

Better logic

Explanation

Inefficient and slow.

11. What happens if you don't close DB connections?

EASY

Leak

Correct Answer

Faster

Compile error

No issue

Explanation

Resource leak.

12. Why use @RestController instead of @Controller?

EASY

Faster

Secure

Less code

JSON response

Correct Answer

Explanation

Automatically returns JSON.

13. What is the issue if API returns sensitive fields like password?

EASY

Performance issue

Security risk

Correct Answer

Memory issue

UI issue

Explanation

Security vulnerability.

14. Why use caching (like Redis)?

EASY

Security

Faster response

Correct Answer

UI

Storage

Explanation

Reduce DB load.

15. What happens if two users update same record simultaneously?

EASY

No issue

Faster

Conflict

Correct Answer

Delete

Explanation

Data conflict.

16. What is the best HTTP method for updating data?

EASY

PUT/PATCH

Correct Answer

GET

POST

DELETE

Explanation

PUT or PATCH

17. Why is GET not used for sensitive data?

EASY

Slow

Not supported

Visible

Correct Answer

Memory

Explanation

Data visible in URL.

18. What is the issue with N+1 query problem?

EASY

Faster

Security

Performance issue

Correct Answer

UI

Explanation

Multiple DB calls reduce performance.

19. What happens if frontend doesn't handle API error?

EASY

UX issue

Correct Answer

Works

Crash

Faster

Explanation

Poor user experience.

20. Why use environment variables in backend?

EASY

Faster

UI

Security

Correct Answer

Memory

Explanation

Secure configuration.

21. How would you reduce API latency in a high-traffic system?

EASY

Add loops

Optimize layers

Correct Answer

Increase CSS

Restart server

Explanation

Use caching, async processing, DB optimization.

22. What is the best way to handle large file upload?

EASY

Store in memory

Streaming

Correct Answer

Ignore

Split UI

Explanation

Streaming instead of loading fully.

23. Why use message queues (Kafka/RabbitMQ)?

EASY

Faster UI

Reduce DB

CSS

Async handling

Correct Answer

Explanation

Async processing.

24. What issue occurs without proper API versioning?

EASY

Faster

Break clients

Correct Answer

Memory

UI

Explanation

Breaking changes for clients.

25. How would you handle authentication in scalable system?

EASY

Sessions only

Cookies only

JWT

Correct Answer

Local storage

Explanation

JWT-based stateless auth.

26. What is the issue with tightly coupled services?

EASY

Hard maintenance

Correct Answer

Faster

Secure

Your Answer

UI

Explanation

Hard to scale/maintain.

27. Why is lazy loading useful in frontend?

EASY

Security

Memory

DB

Faster load

Correct Answer

Explanation

Improves load performance.

28. What is optimistic locking?

EASY

Lock DB

Delete data

Version check

Correct Answer

Cache

Explanation

Prevents conflicts using versioning.

29. Why avoid long-running DB transactions?

EASY

Lock issues

Correct Answer

Faster

UI issue

Memory

Explanation

Locks resources and reduces performance.

30. What is the best approach to debug production issue?

EASY

Restart

Delete code

Ignore

Logs & monitoring

Correct Answer

Explanation

Logs + monitoring + tracing.